

Confining a Language Model to a Prescribed Role: An Adversarial Reliability Test for Instruction Chatbots in Economics Experiments

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In many economics experiments a research assistant answers participants’ questions about the task. The assistant must explain the rules, but must never give advice or hint at the best choice—that would change the very behavior being studied. We ask a simple question: can a language-model chatbot be trusted to stay inside this role, and what makes it reliable? We build such a chatbot for Oprea’s (2024) experiments on choice under risk, where the central debate is about separating what participants *understand* from how *complex* the task is to think through. Our chatbot is therefore allowed to clarify the rules, but is forbidden to give advice, work out the best option, or make the task look simpler than it is. To test it without human participants, we let adversarial “synthetic subjects”—language models playing seven kinds of difficult participant—try to pull the chatbot out of its role over many 15-turn conversations, more than 27,000 replies in total. An independent model then judges every reply: did it break the no-advice rule, say something false, or fail to make sense? Using this setup we compare different chatbot designs.

The main result is clear. A *simpler* design is safer: a single, well-instructed model call is the most reliable, and every extra “guardrail” agent we add makes things worse, not better, because the added steps tend to repeat the participant’s forbidden question back to them. The one change that helps is a *stronger base model*: it cut rule-breaking replies from about 12% to 4% on the hardest test, while also being more accurate and faster. Adding reasoning steps or a correction pass did not help. We measure reliability over a whole conversation, not a single reply, because small per-reply risks add up: even a chatbot that slips only 4% of the time will slip at least once in most 15-turn conversations. These results are still preliminary, but the lesson is already clear—for a chatbot that must stay in a strict role, the quality of the base model matters far more than how elaborate the surrounding workflow is. Full methods and interactive results: can-celebi.github.io/chatbot-for-experiments.